

CAPSTONE PROJECT 1

Title: Predict Potential Spammers on Fiverr Platform

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1. PROBLEM STATEMENT:

Attackers use Fiverr to distribute malware like Agent Tesla via fake job offers. This causes users to lose earnings and account access. This project builds an ML model to predict potential spammer accounts before they cause financial harm.

2. OBJECTIVE:

To develop a classification model using anonymized user behavior features X1-X51 to identify potential spammer accounts on Fiverr with high recall for the spammer class.

3. DATASET DESCRIPTION:

Source: Kaggle Competition - Predict Potential Spammers on Fiverr

Size: 458,798 rows, 51 behavioral features + user_id + label

Target: label (1 = Spammer, 0 = Genuine User)

Class Distribution: 97.31% Genuine (446,477), 2.68% Spammer (12,321) - Highly Imbalanced

4. METHODOLOGY:

4.1 Preprocessing: Dropped user_id. Used 80-20 stratified train-test split. Test set: 91,760 users.

4.2 Algorithm: Random Forest Classifier with n_estimators=100 and class_weight='balanced'

to penalize misclassification of the minority spammer class.

4.3 Reason: Random Forest handles high-dimensional tabular data, reduces overfitting, and

provides feature importance crucial for cybersecurity interpretability.

5. RESULTS:

5.1 Accuracy: 98.52% - High due to class imbalance.

5.2 ROC-AUC Score: [Yaha apna ROC-AUC number likh do, jaise 0.9015]

5.3 Classification Report:

Genuine: Precision 0.99, Recall 1.00, F1-Score 0.99

Spammer: Precision 0.83, Recall 0.56, F1-Score 0.67

5.4 Confusion Matrix: [[89018, 278], [1084, 1380]]

- True Negatives: 89,018 Genuine users correctly identified.
- False Positives: 278 Genuine users incorrectly flagged. Very low FP rate.
- False Negatives: 1,084 Spammers missed.
- True Positives: 1,380 Spammers correctly caught.

5.5 Key Finding: Model correctly identifies 1,380 out of 2,464 spammers with 83% precision,

while only false-flagging 278 genuine users out of 89,296.

5.6 Feature Importance: Top 3 predictive features are X19 (0.137), X6 (0.067), and X7 (0.061).

6. CONCLUSION & BUSINESS IMPACT:

The model achieves 56% Spammer Recall with 83% Precision. The low False Positive count of

278 ensures minimal disruption to genuine users. By flagging 1,380 high-risk accounts, Fiverr can prevent significant malware distribution.

Limitation: Anonymized features restrict domain insights.

Future Scope: Deploy as real-time API to screen new user signups.

7. TECH STACK: Python, Pandas, Scikit-learn, Matplotlib, Seaborn, Google Colab